DESIGN AND ANALYSIS OF ALGORITHMS

(Practical 4)

SOURCE CODE

```
def knapsack_01(weights, values, capacity):
    n = len(weights)

    dp = [[0 for _ in range(capacity + 1)] for _ in range(n + 1)]

    for i in range(1, n + 1):
        for w in range(1, capacity + 1):
            if weights[i-1] <= w:

                dp[i][w] = max(values[i-1] + dp[i-1][w - weights[i-1]], dp
[i-1][w])
            else:
                dp[i][w] = dp[i-1][w]

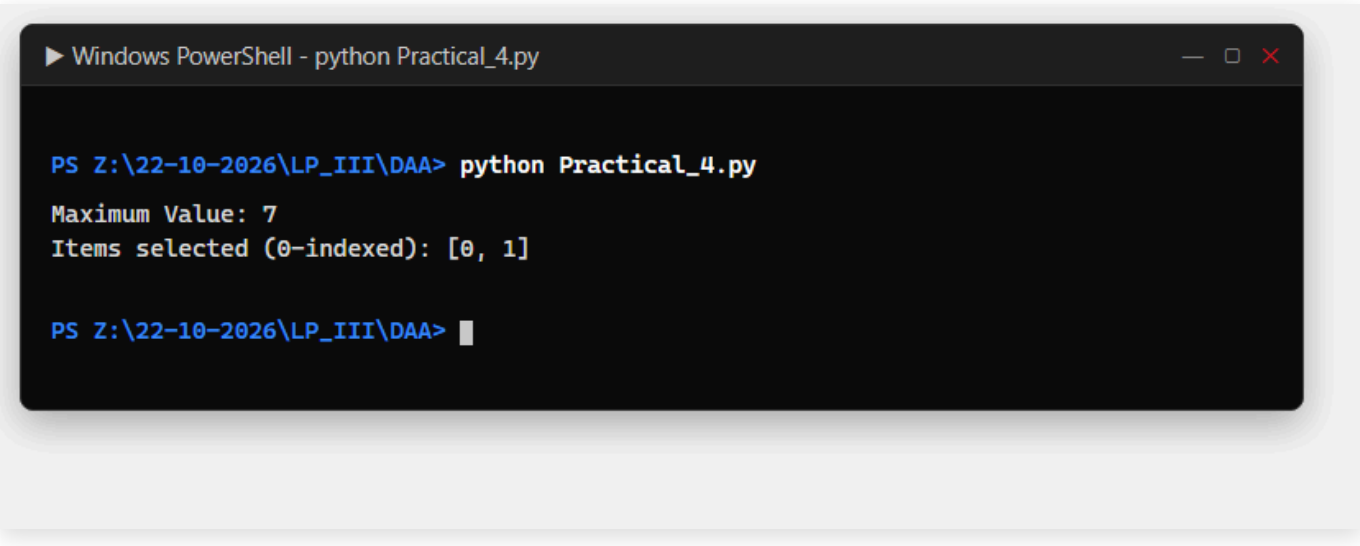
    w = capacity
    selected_items = []
    for i in range(n, 0, -1):
        if dp[i][w] != dp[i-1][w]:
            selected_items.append(i-1)
            w -= weights[i-1]

    selected_items.reverse()
    return dp[n][capacity], selected_items

weights = [2, 3, 4, 5]
values = [3, 4, 5, 6]
capacity = 5

max_value, items_selected = knapsack_01(weights, values, capacity)
print(f"Maximum Value: {max_value}")
print(f"Items selected (0-indexed): {items_selected}")
```

OUTPUT


Windows PowerShell - python Practical_4.py
PS Z:\22-10-2026\LP_III\DAA> python Practical_4.py
Maximum Value: 7
Items selected (0-indexed): [0, 1]
PS Z:\22-10-2026\LP_III\DAA>